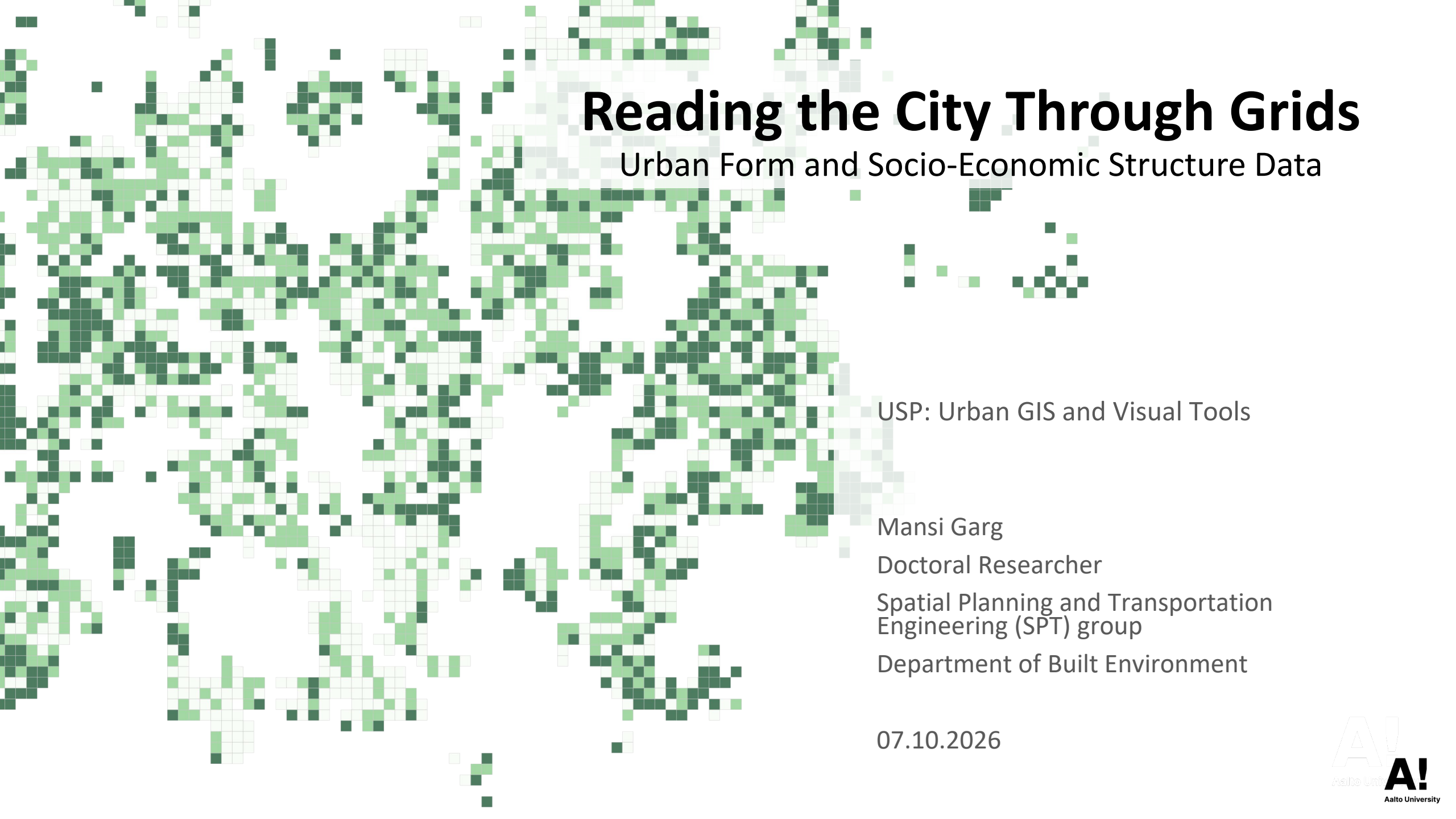




Reading the City Through Grids

Urban Form and Socio-Economic Structure Data

USP: Urban GIS and Visual Tools

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Department of Built Environment

07.10.2026

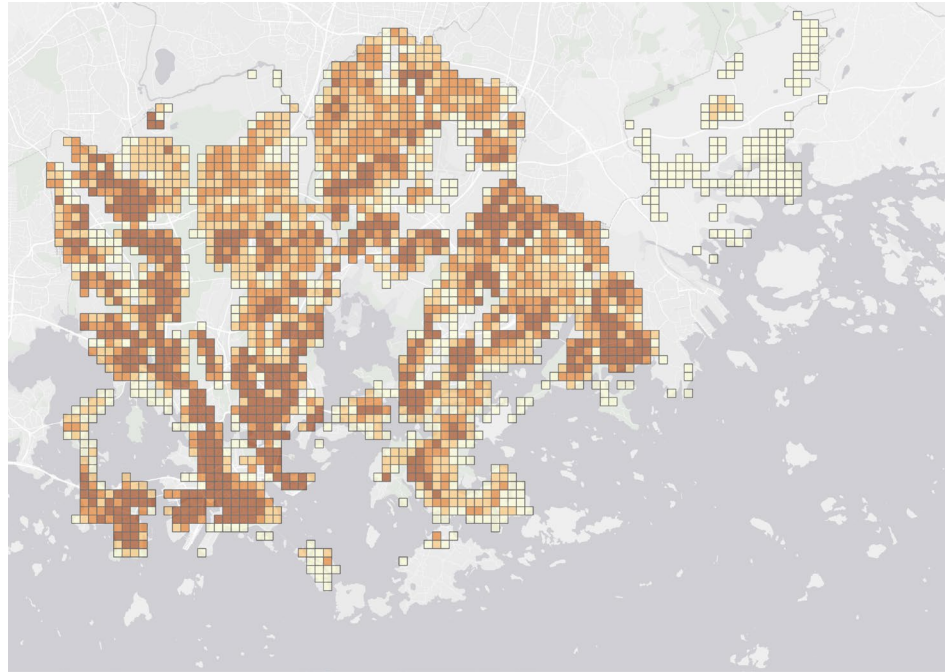
Content

- Understanding urban structure grid data
- Applications of urban structure grid data
- Best practices to visualize grid data

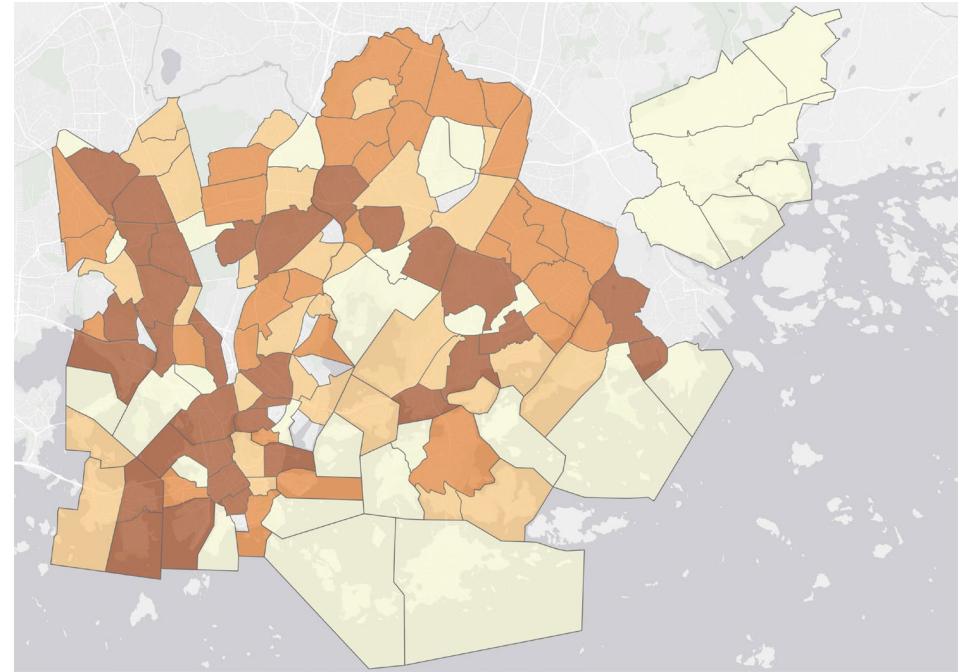
Quick warm up

A city wants to add new bicycle parking stands near areas with high local demand.

The following maps represent population density on two different spatial units. Which is better for deciding bicycle parking locations?

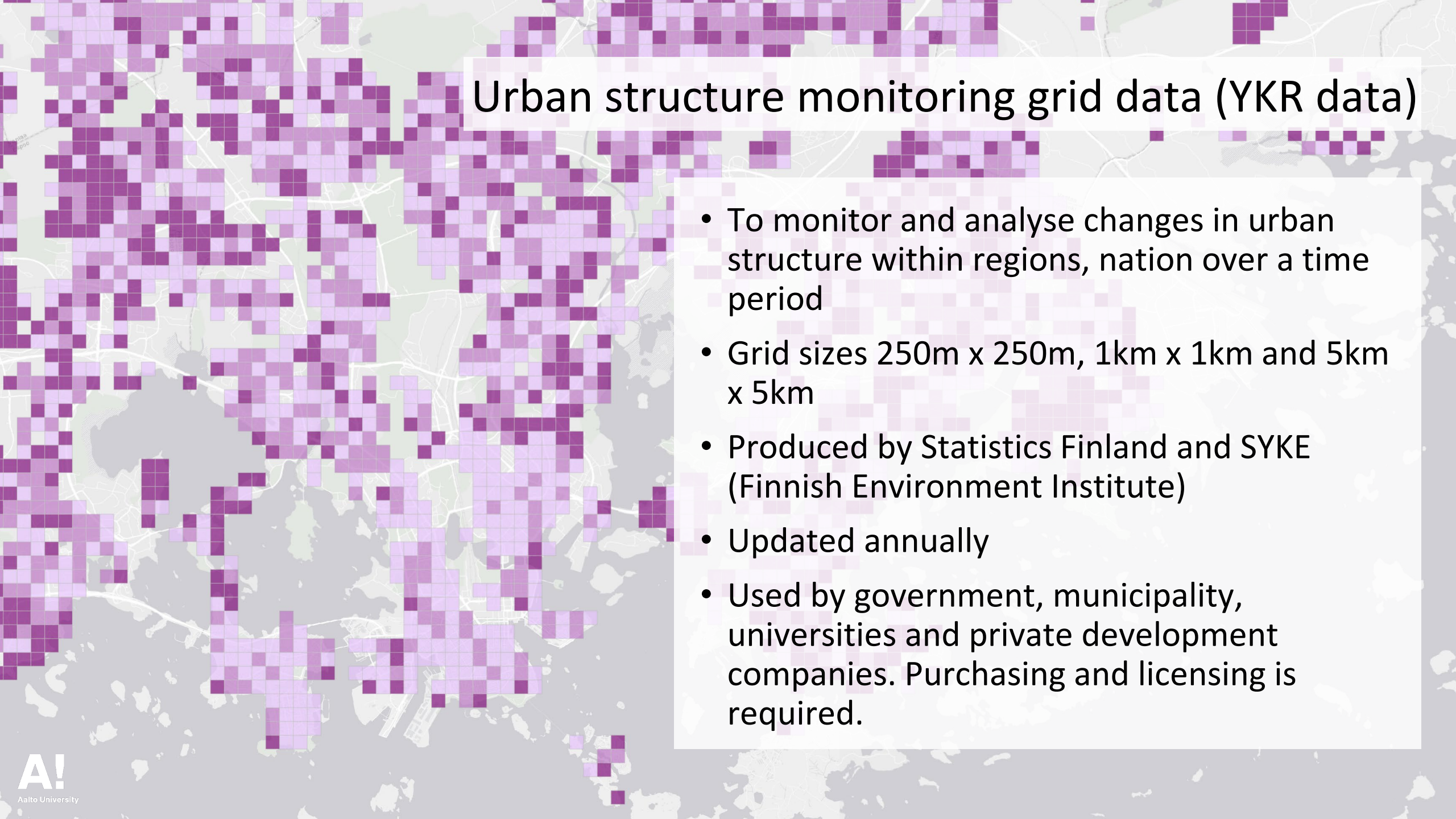


Grid-based population density map



District boundary map

Maps are for representative purposes only and do not show actual population distribution.



Urban structure monitoring grid data (YKR data)

- To monitor and analyse changes in urban structure within regions, nation over a time period
- Grid sizes 250m x 250m, 1km x 1km and 5km x 5km
- Produced by Statistics Finland and SYKE (Finnish Environment Institute)
- Updated annually
- Used by government, municipality, universities and private development companies. Purchasing and licensing is required.

Data groups in YKR data

Kunta (Municipality Code)

Helsinki 091

Espoo 049

Vantaa 092

Population structure (HE data group code; var names: he_vakiy, he_kika and so on..)

Educational structure (KO)

Inhabitants' disposable monetary income (HR)

Size and stage in life of households (TE)

Households' disposable monetary income (TR)

Buildings and housing (RA)

Workplace structure (TP)

Main type of activity (PT)

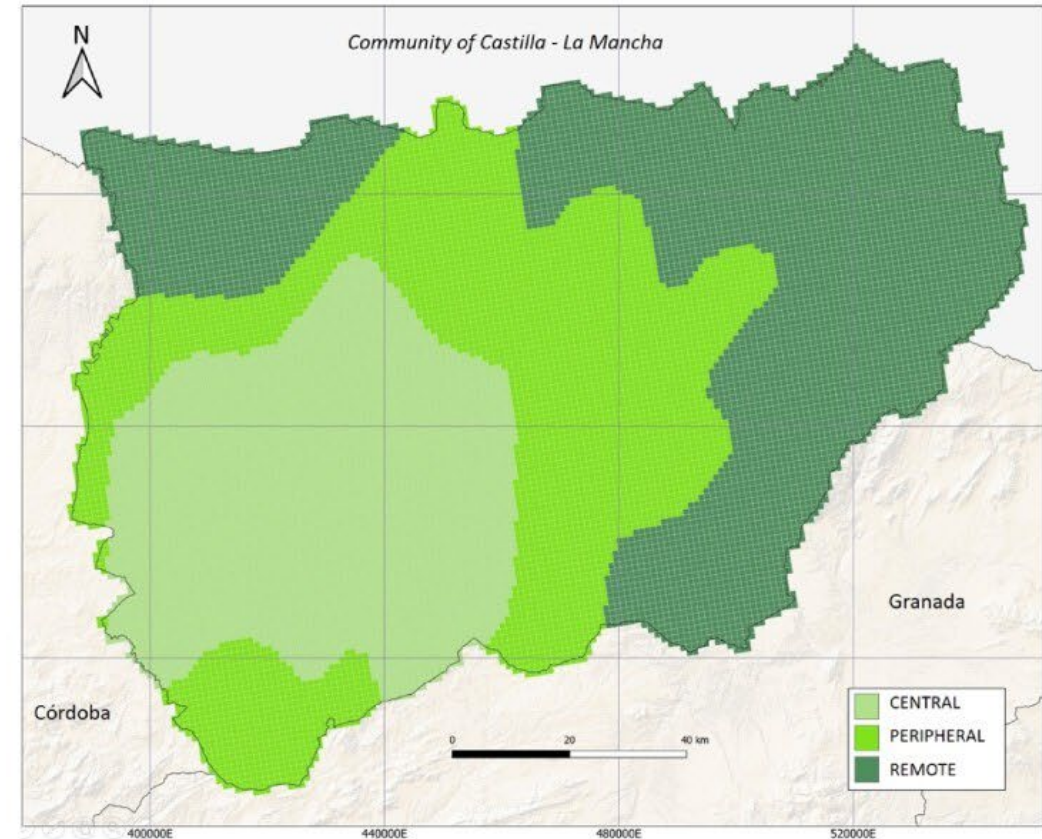
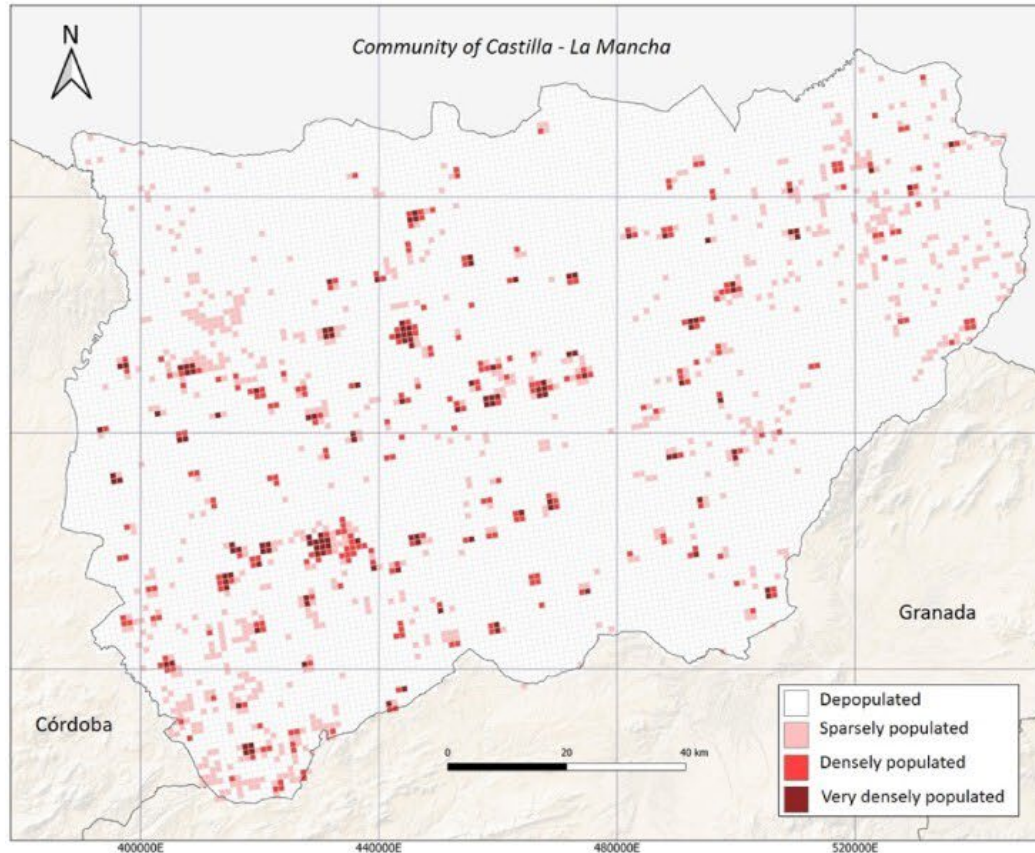
<

Quick reflection

For example, we take variable pt_elakel (Pensioners) from data group 'Main type of activity'. What happens when a grid has less than 10 cases?

pt_opisk	pt_elakel	pt_muut
1	4	0
0	0	0
-1	-1	-1
0	0	0
0	0	0
0	0	0
0	0	0
-1	-1	-1

Applications of urban structure grid data



Population distribution and accessibility to the provincial capital by road (2021).

Use of statistical grid data to delimit and classify rural and urban areas beyond traditional municipality-based boundaries.

Sources: Instituto de Estadística y Cartografía de Andalucía (IECA) y HERE Maps. Sánchez-Martínez, J. D., & Garrido-Almonacid, A. (2024). Delimitation of rural and urban areas from spatial data in statistical grids: the province of Jaén (Spain) as a case study. *Journal of Maps*, 20(1). <https://doi.org/10.1080/17445647.2024.2301980>

Applications of urban structure grid data

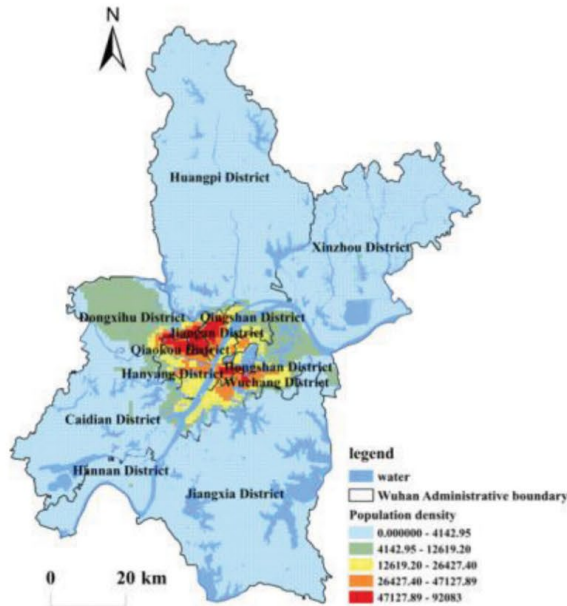


Figure.2 Distribution of population data

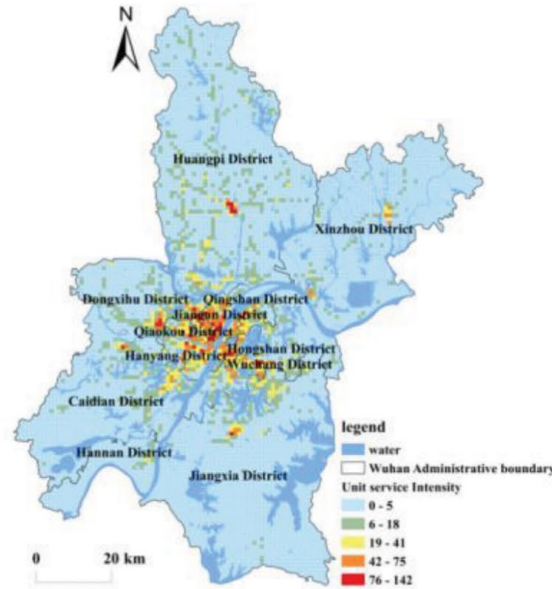


Figure 3 Public transport service configuration distribution

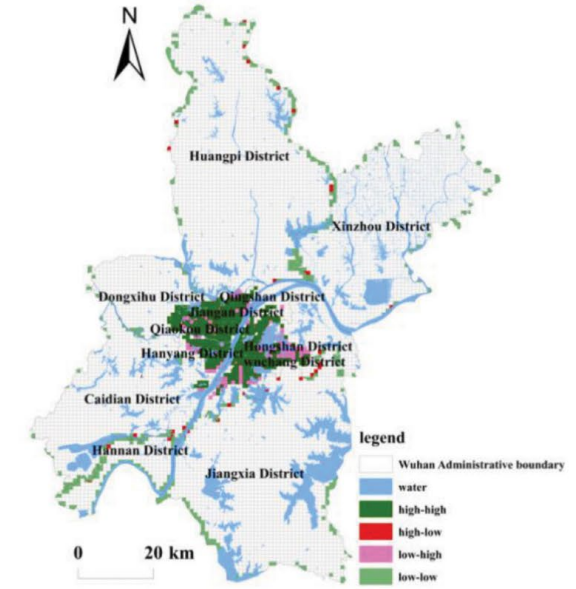


Figure. 4 Evaluation result at the local scale

Evaluating the spatial fairness of public transport services in Wuhan by using grid-based spatial units to compare transport supply and population/service demand across the city.

Applications of urban structure grid data

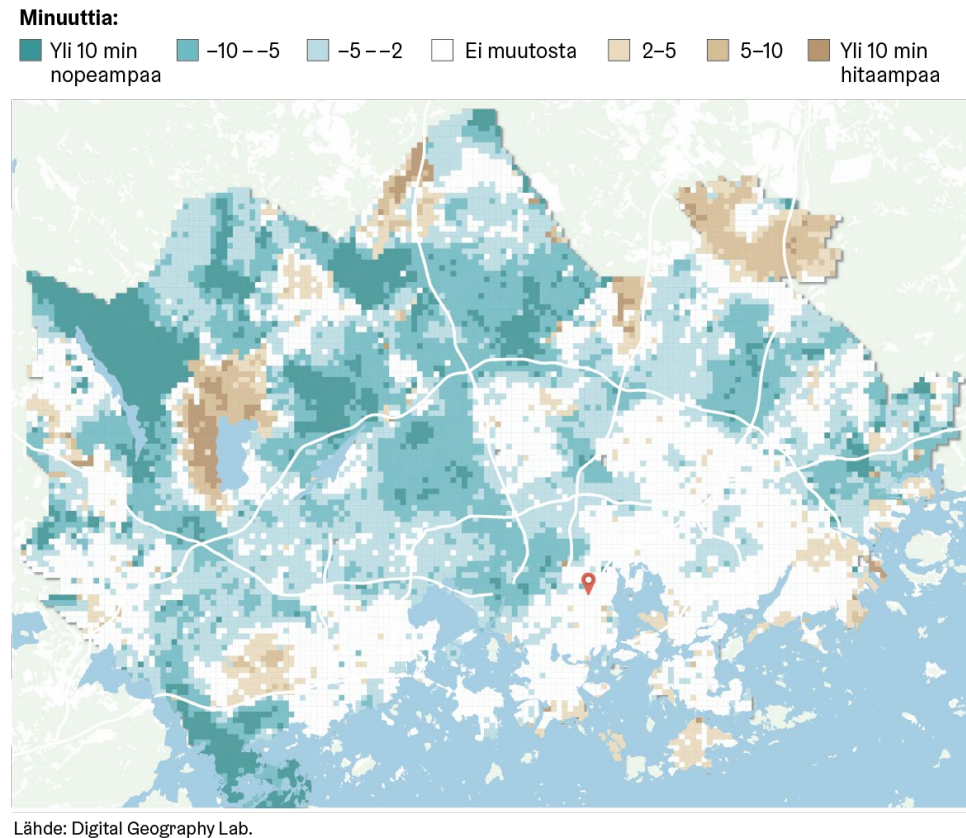


Figure 3. Change in public transport travel times to Kumpula Campus (2018–2023)

“When looking at the change in public transport accessibility to the Kumpula campus from 2018 to 2023, we see that travel times have increased slightly on average.”

<https://www.helsinki.fi/fi/tutkimusryhmat/digital-geography-lab/paakaupunkiseudun-matka-aikamatriisi-2023#julkinen-liikenne>

<https://www.helsinki.fi/fi/tutkimusryhmat/digital-geography-lab/paakaupunkiseudun-matka-aikamatriisi-2023>

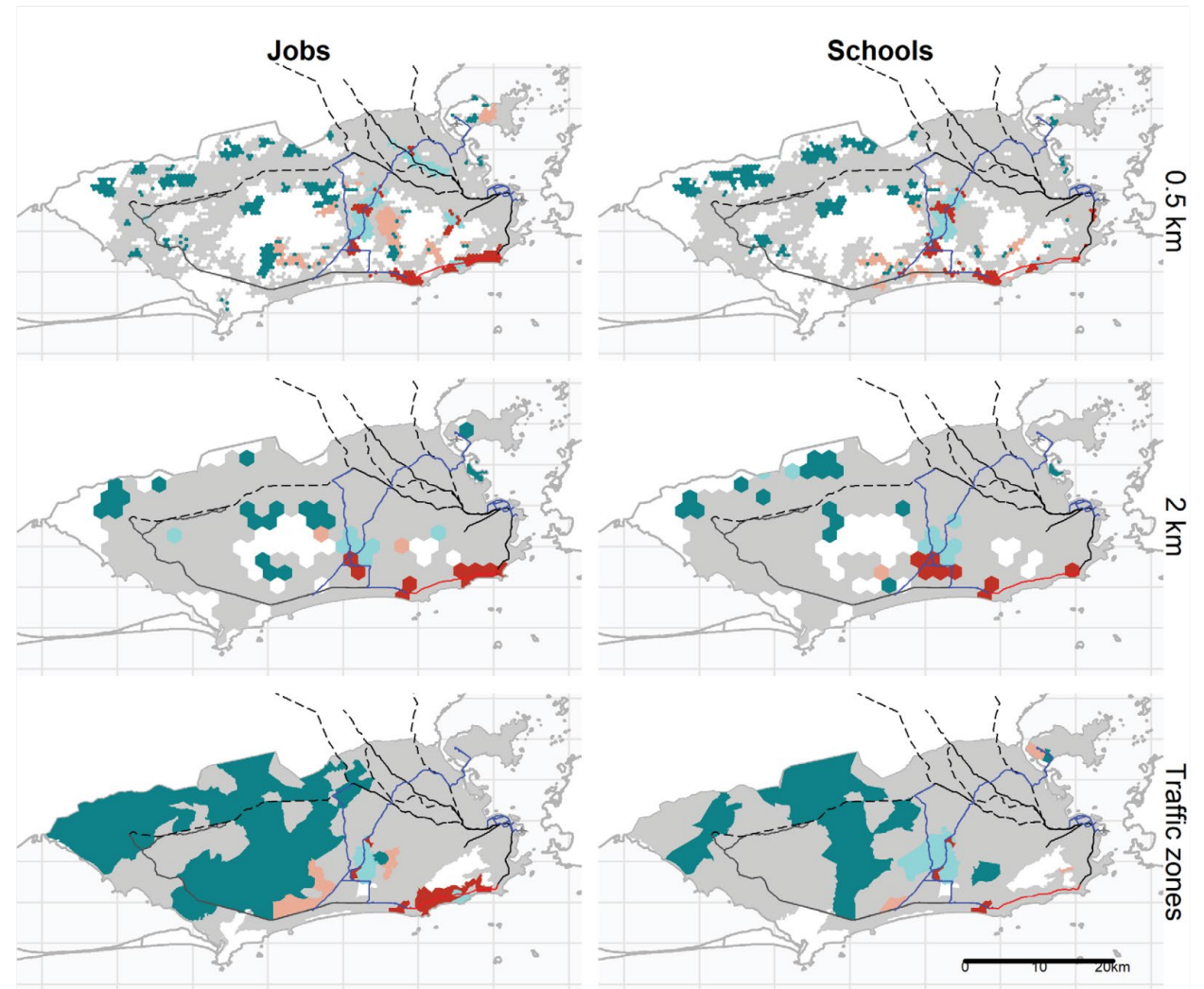
Applications of urban structure grid data

Fine-scale spatial units/grid-based accessibility analysis to estimate how many jobs and opportunities residents in different parts of Rio de Janeiro could reach by public transport under different policy scenarios.

LISA clusters

- High income - High access gain
- High income - Low access gain
- Low income - High access gain
- Low income - Low access gain
- Not significant

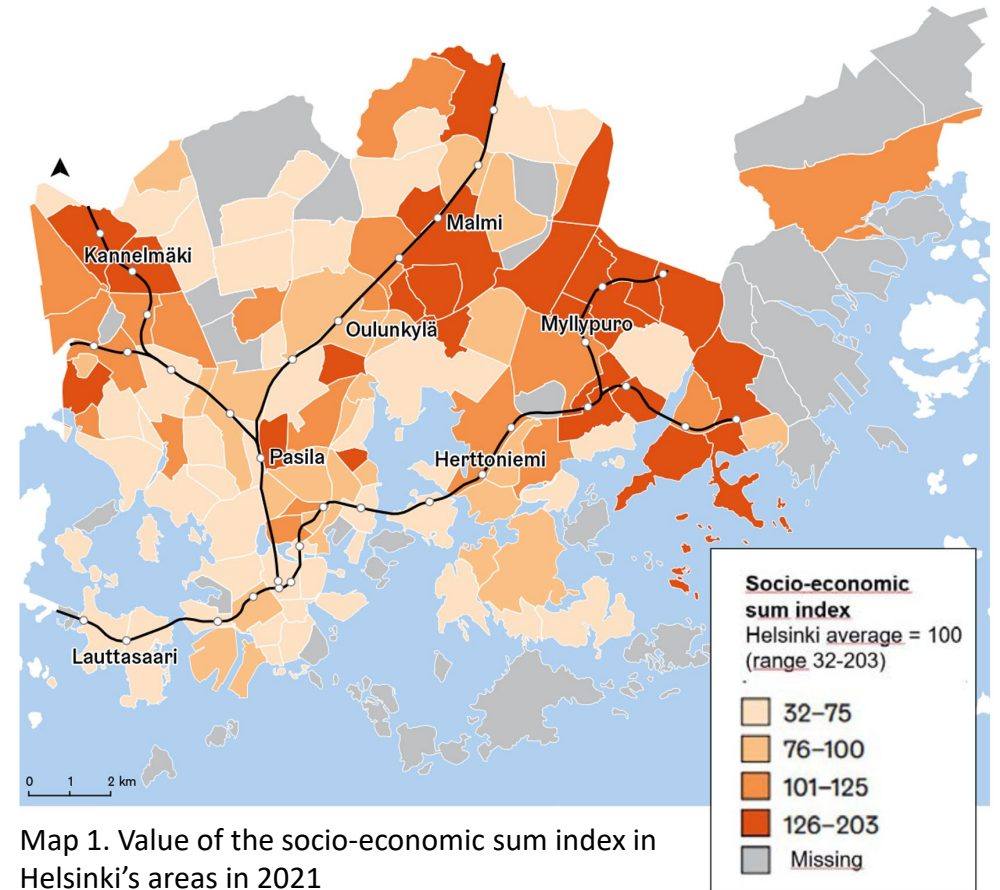
Figure 6. Bivariate LISA based spatial clusters showing the local association between average household income per capita and gains in access to jobs and schools by public transport between 2014 and 2017, Rio de Janeiro



Today's practical : Socio-economic structure

Socio-Economic Sum index (SES index)

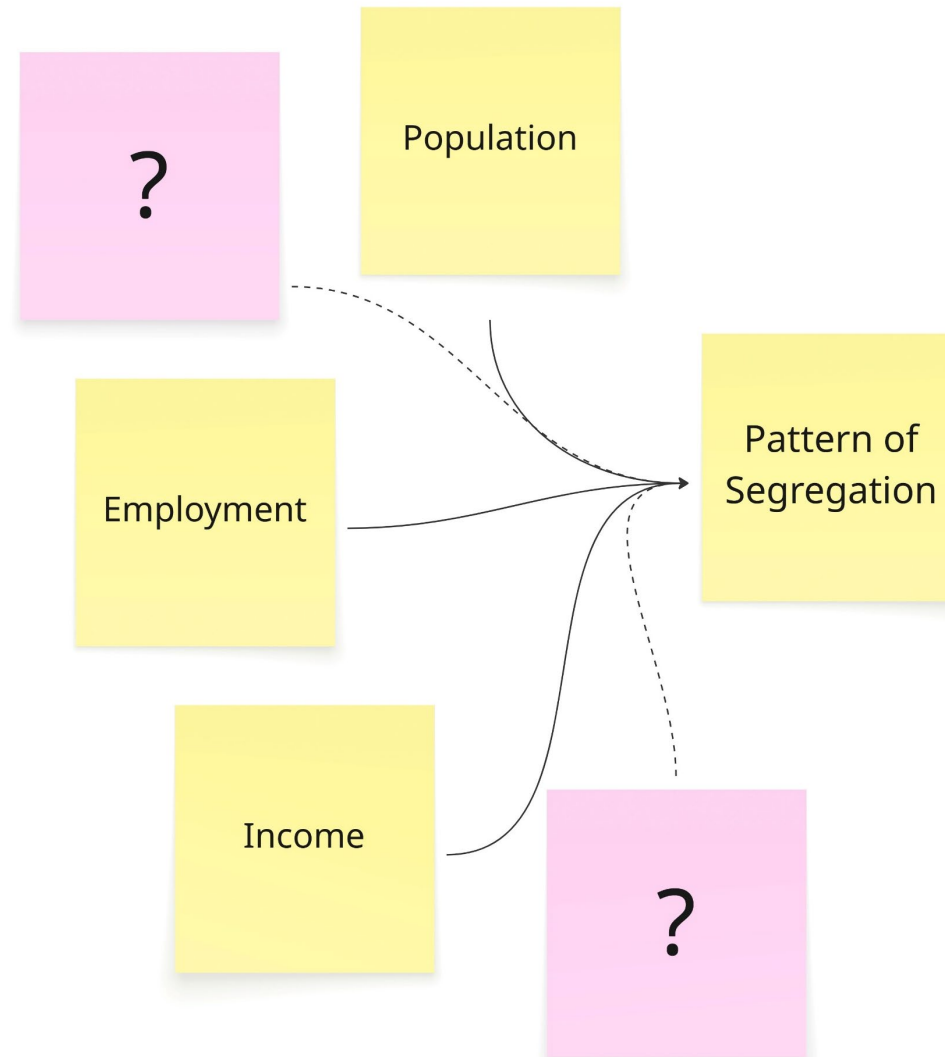
combines unemployment, low education, and low-income shares relative to the city average



Map 1. Value of the socio-economic sum index in Helsinki's areas in 2021

Today's practical : Socio-economic structure

Patterns of
Segregation using
YKR grid data



Brainstorming Activity (5mins)

Discuss with the person next to you and identify **one application of grid data analysis** beyond the examples shown in class.

Focus on:

- What problem or question would you analyse?
- What kind of data do you need?
- How could the results support planning or policy decisions?

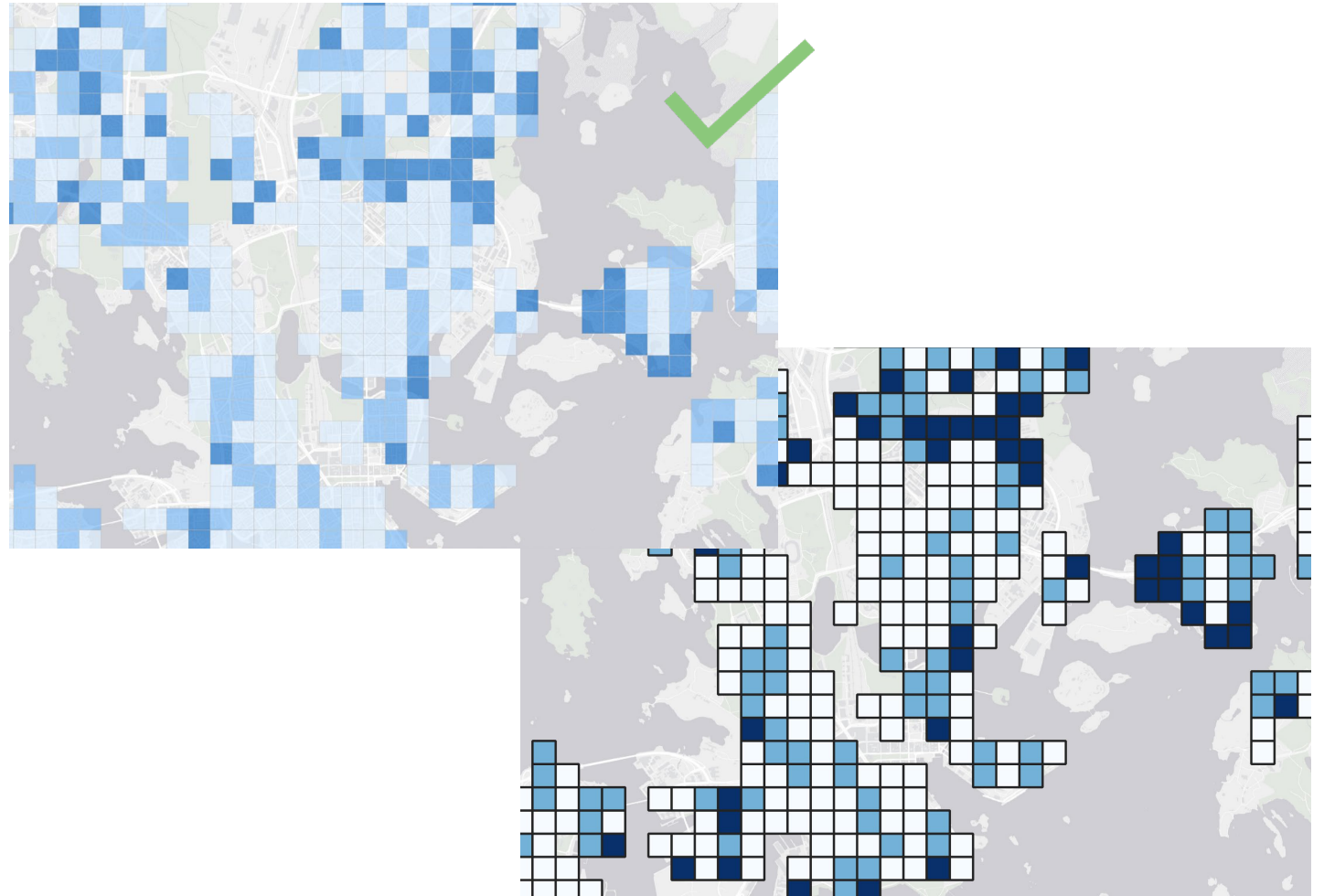
Best practices to visualize grid data maps

Check

- Color and the width (thickness) of gridlines

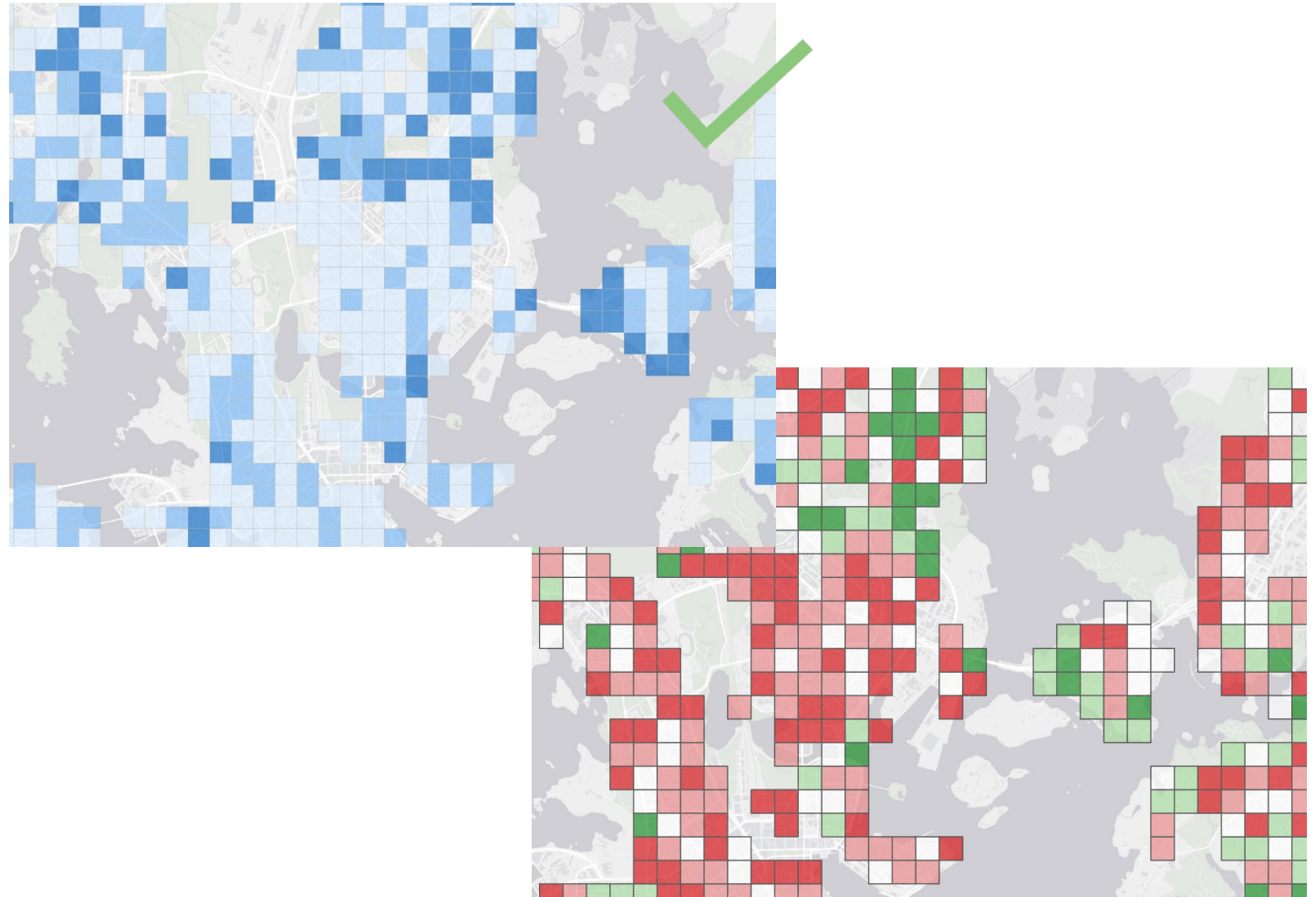
Prefer gray or subdued colors, desaturated colors to avoid cluttering the data visualization and maintains enough contrast to be visible.

- Transparency of the grid or map



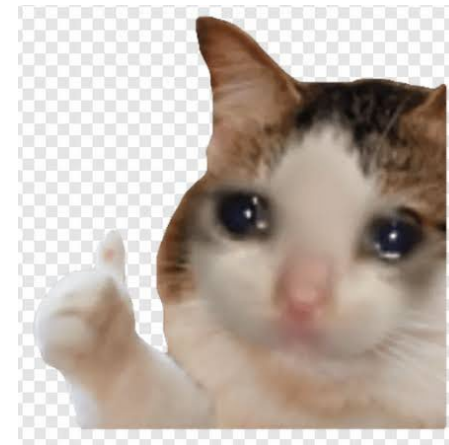
Best practices to visualize grid data maps

- Choose appropriate colors for the grid cells
 - Use intuitive colors (water – blue, heat – red etc.)
 - Use light colors for low values and dark colors for high values.
 - Avoid green and red gradients together for accessibility
 - Avoid gradient color palette for categories and similarly avoid multiple colors for high and low values. (Grids are mostly high and low values)
- Legends
(Don't forget the key 🗝️ !)
- North, Scale, Title, Data and Basemap Source

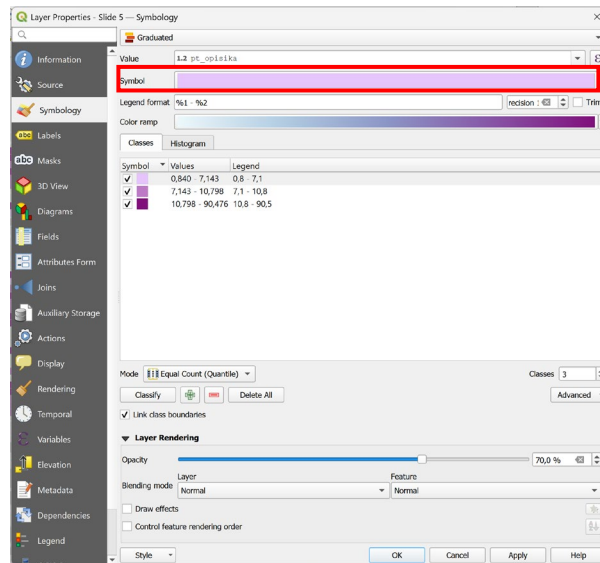


Some useful sources

- Vantaa map service <https://kartta.vantaa.fi>, you can possibly extract everything there:
 - raster maps : masterplans, local detailed plans
 - building typology, building construction year
 - Road networks, bicycle corridors
 - Services in the areas (Schools, hospital, daycares, community buildings)
- Similarly <https://kartat.espoo.fi/IMS/en/Map> for Espoo and <https://kartta.hel.fi/?setlanguage=en> for Helsinki
- WMS and WFS layers for Helsinki <https://www.hel.fi/fi/paatoksenteke-ja-hallinto/tietoa-helsingista/kartat-ja-paikkatieto/paikkatiedot-tehokayttoon/avoimet-paikkatiedot> and <https://kartat.espoo.fi/avoindata/> for Espoo
- Open data from HSL <https://public-transport-hslhrt.opendata.arcgis.com/>
 - Bus stops
 - Bus routes
 - Ridership figures
- Helsinki Region Infoshare <https://hri.fi/fi/>
- **Statistics Finland Database** <https://pxdata.stat.fi/PxWeb/pxweb/en/StatFin/>
 - Energy consumption, prices and even energy production from various sources
 - Road statistics
 - Housing conditions etc.

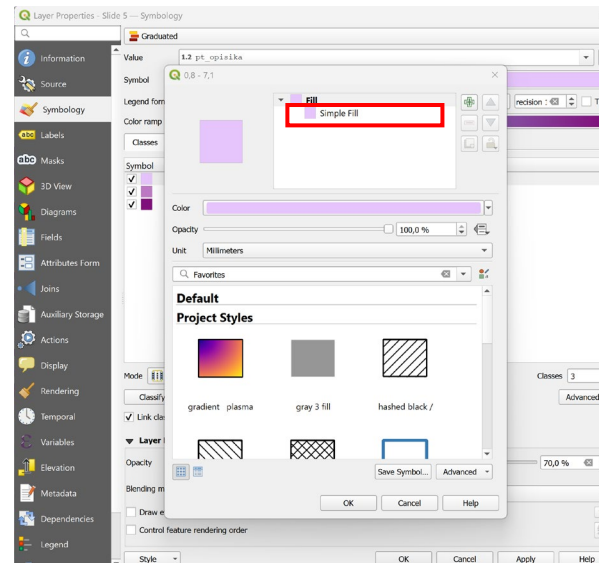


Bonus: How to customise grid lines



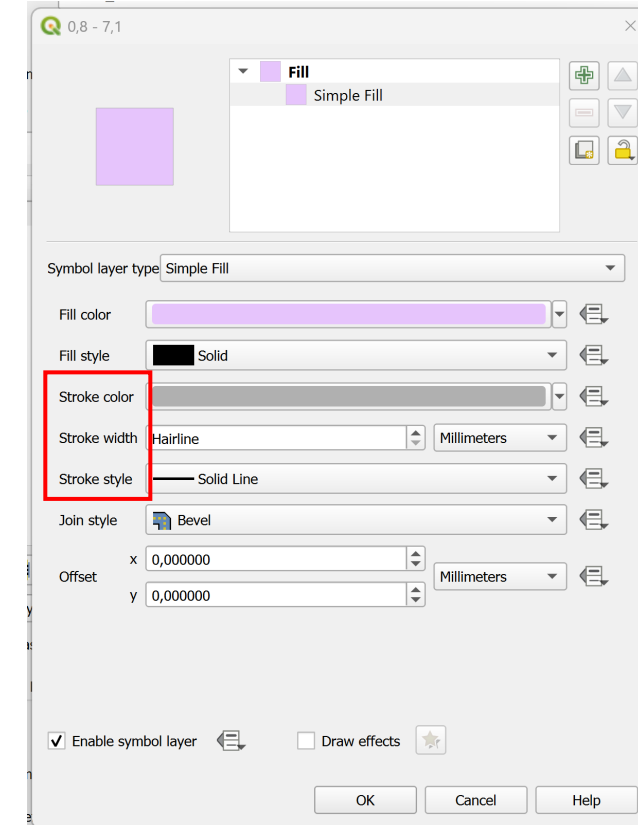
Right clicking on your layer to open properties > go to symbology > click the symbol tab

It will open a new dialogue box



Click Simple fill

It will also open a new dialogue box



Edit Stroke width, color and style to customise your grid lines.